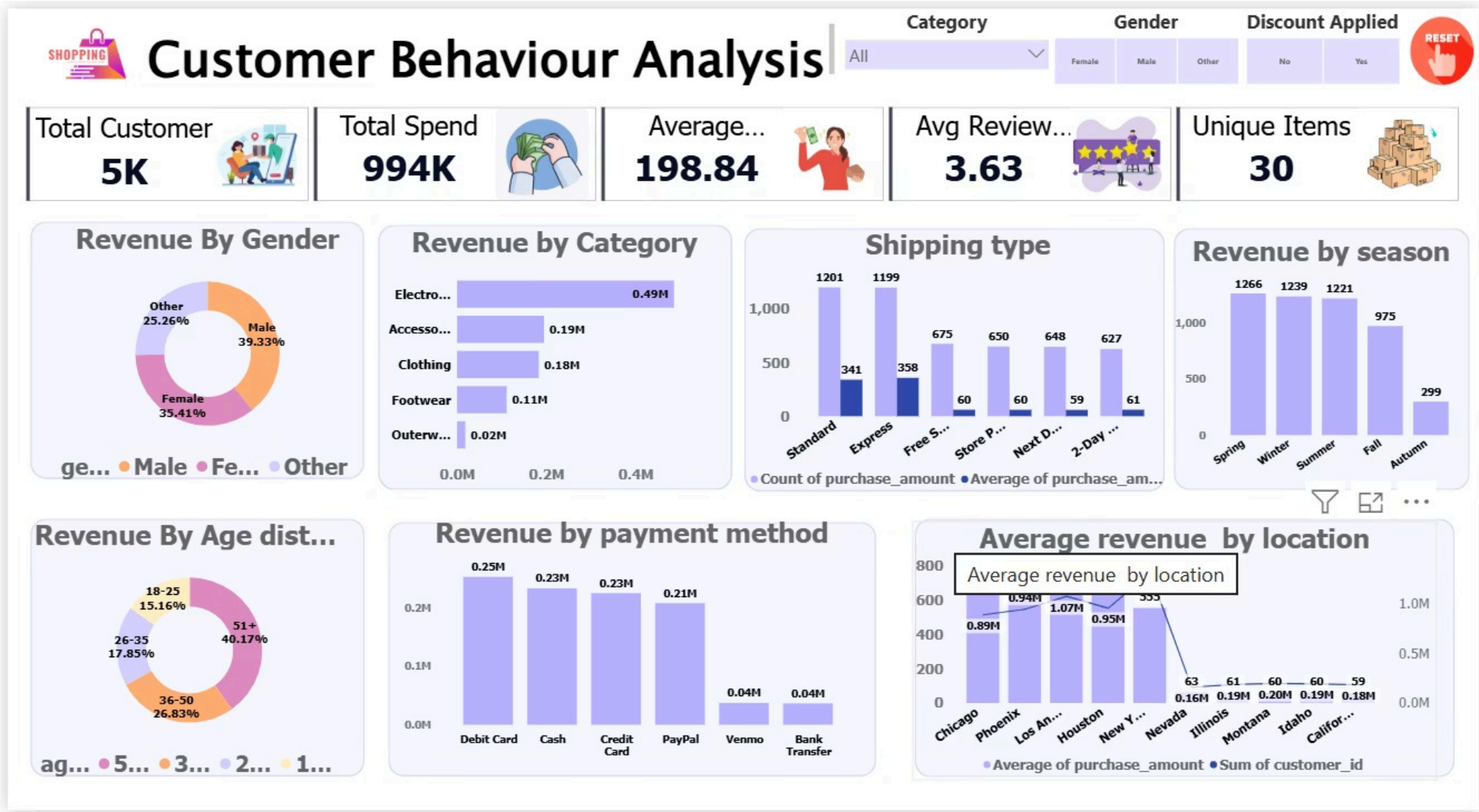


Customer Behaviour Analysis

Retail / E-commerce dataset – customer transaction level data for behavior analysis & business decision-making.

Numerical Age, Purchase Amount, Previous Purchases	Categorical Category, Payment Method, Season	Level Customer Transaction Level
--	--	--



Why This Data Matters



Business Goals

- Identify high-value customers
- Understand purchase patterns
- Optimize discount strategies
- Increase Customer Lifetime Value (CLV)



The Business Problem

How can a retail company understand purchasing behavior to **increase revenue, improve retention, and optimize marketing?**

Low Retention

Who are frequent vs. one-time buyers?

Ineffective Campaigns

Do discounts actually drive purchases?

Seasonal Demand

Which season drives maximum sales?

Segmentation

High-value, frequent, discount-sensitive customers

Data Dictionary

```
... <class 'pandas.DataFrame'>
RangeIndex: 5050 entries, 0 to 5049
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Customer ID           5050 non-null   int64
 1   Age                   5050 non-null   int64
 2   Gender                5050 non-null   str
 3   Item Purchased        5050 non-null   str
 4   Category              5050 non-null   str
 5   Purchase Amount (USD) 4494 non-null   float64
 6   Location              5050 non-null   str
 7   Size                  4680 non-null   str
 8   Color                 5050 non-null   str
 9   Season                5050 non-null   str
10   Review Rating         4449 non-null   float64
11   Subscription Status    5050 non-null   str
12   Shipping Type         5050 non-null   str
13   Discount Applied      5050 non-null   str
14   Previous Purchases    4502 non-null   float64
15   Payment Method        5050 non-null   str
16   Frequency of Purchases 5050 non-null   str
dtypes: float64(3), int64(2), str(12)
memory usage: 670.8 KB
```

18 columns covering customer demographics, product details, and transaction behavior.

Column	Description
Customer ID	Unique identifier
Age / Gender	Customer demographics
Category	Clothing, Electronics, etc.
Purchase Amount	Amount spent (USD)
Season	Winter, Summer, Spring, Fall
Discount Applied	Yes / No
Previous Purchases	Number of past purchases
Subscription Status	Yes / No
Payment Method	Cash, Card, PayPal, etc.
Frequency of Purchases	Weekly, Monthly, etc.

EDA: Data Cleaning in Python

Data Cleaning

```
df.head()
```

```
df['Item Purchased']
```

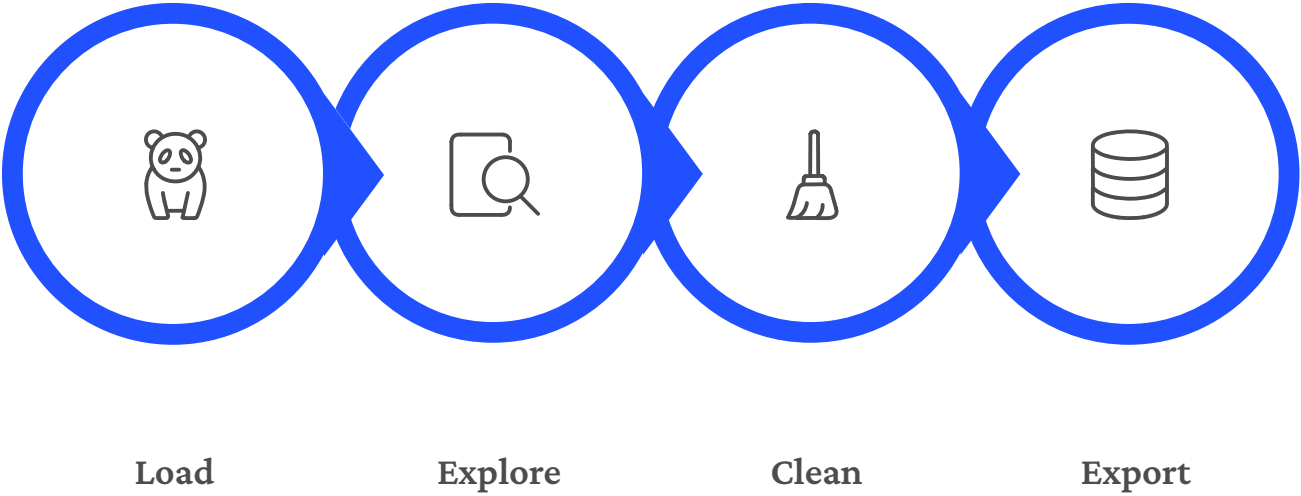
```
df['Item Purchased'].unique()
```

```
df['category'].unique()
```

```
df.groupby("category") ["Item Purchased"].unique()
```

```
df.groupby("category") ["Item Purchased"].unique()
```

```
#maximize column width
pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_column', None)
pd.set_option('display.max_rows', None)
```



Key Cleaning Decisions

Category Correction

Mapped items to correct categories via dictionary

Missing Size

Electronics/Accessories → "Not Applicable"; Clothing → mode

Review Rating

Filled with product-level mean (advanced imputation)

Previous Purchases

Missing = 0 (no prior purchase)

SQL Analysis: Revenue Insights

Object Explorer

Connect

localhost (SQL Server 17.0.1000.7 - AnkitPal\ankit)

Databases

System Databases

Database Snapshots

Cb_Analysis

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.customer_data

Dropped Ledger Tables

Views

External Resources

Synonyms

Programmability

Query Store

Regressed Queries

Overall Resource Consumption

Top Resource Consuming Queries

Queries With Forced Plans

Queries With High Variation

Query Wait Statistics

Tracked Queries

Service Broker

Storage

Security

Financial

MyProject_db

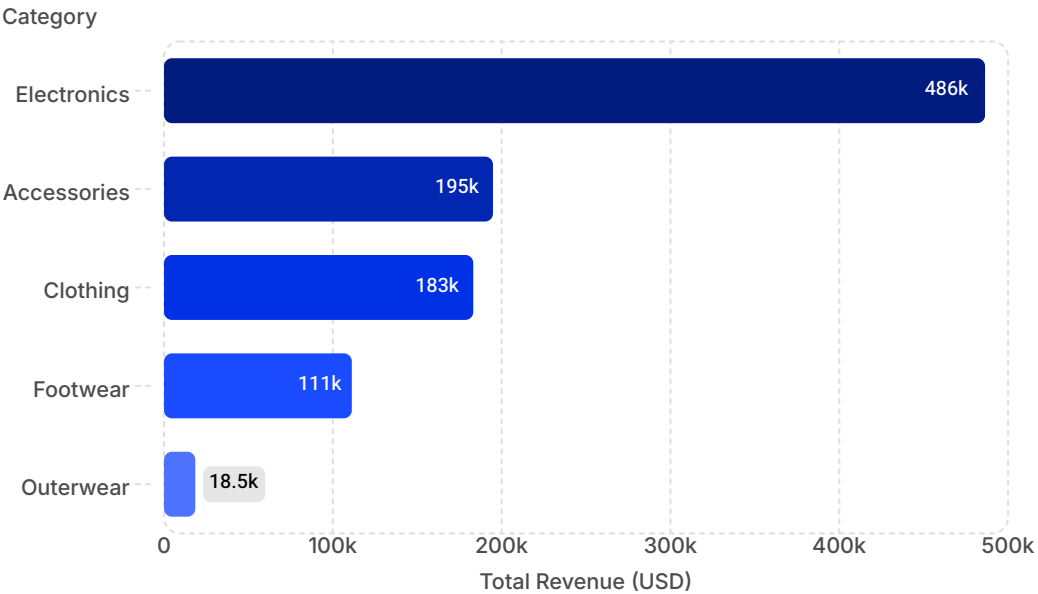
SQLQuery1.s... \ankit (90)

SQLQuery1.sql... L\ankit (58)

```
4
5  -----
6  -- Business Insight Question
7  -----
8  --1. Which Category generates highest Revenue
9  Select
10     category,
11     Round(Sum(purchase_amount),2) as highest_revenue
12   From [dbo].[customer_data]
13   Group by category
14   Order by highest_revenue desc;
15
16  --?.Business problem:- Company does not know which category most to revenue.
17  -- impact -
18      -- Helps prioritize high performing categories
19      -- optimizes inventory planning
20      -- Improves marketing ROI
21  -----
22  --2.Are Discount actually increases Purchase value
23  Select * From customer_data
24
25  Select discount_applied,
26         Round(Sum(purchase_amount),2) as Total_revenue,
27         Round(Avg(purchase_amount),2) as Avg_Purchase
28   From customer_data
29   group by discount_applied;
30
31  --?.Business Problem:- Discount may Reduce Profit without increasing sales
32
33  --Impact
34      --Identify effectiveness of Discount
35      --Reduce Unnecessary discount cost
36      --Improve profit margin
```

98 % 4 0 Ln: 17, Ch: 58 SPC CRLF UTF-8

Connected. (1/1) localhost (17.0 RTM) ANKITPAL\ankit (90) Cb_Analysis 00:00:00 Row: 0, Col: 0 0 rows



Key Findings

Electronics #1

\$486K – top revenue category by far

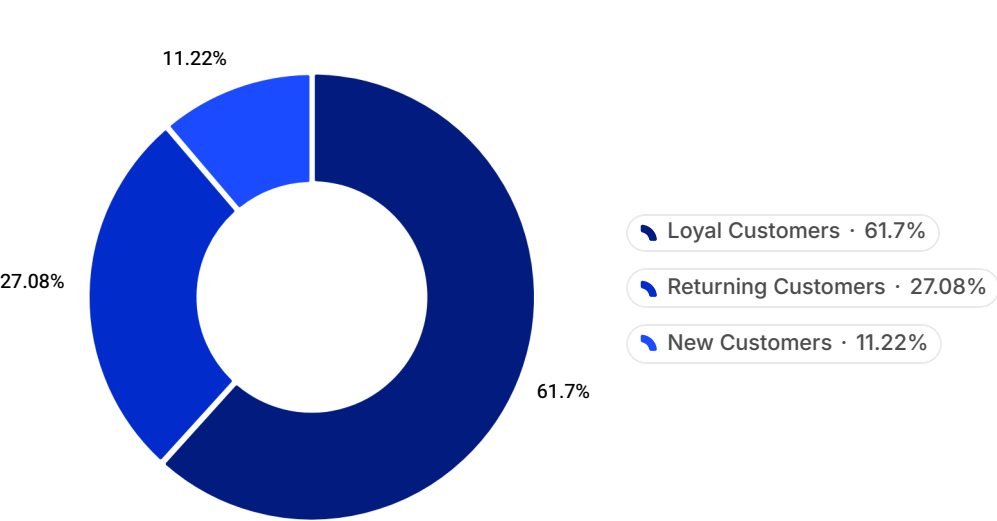
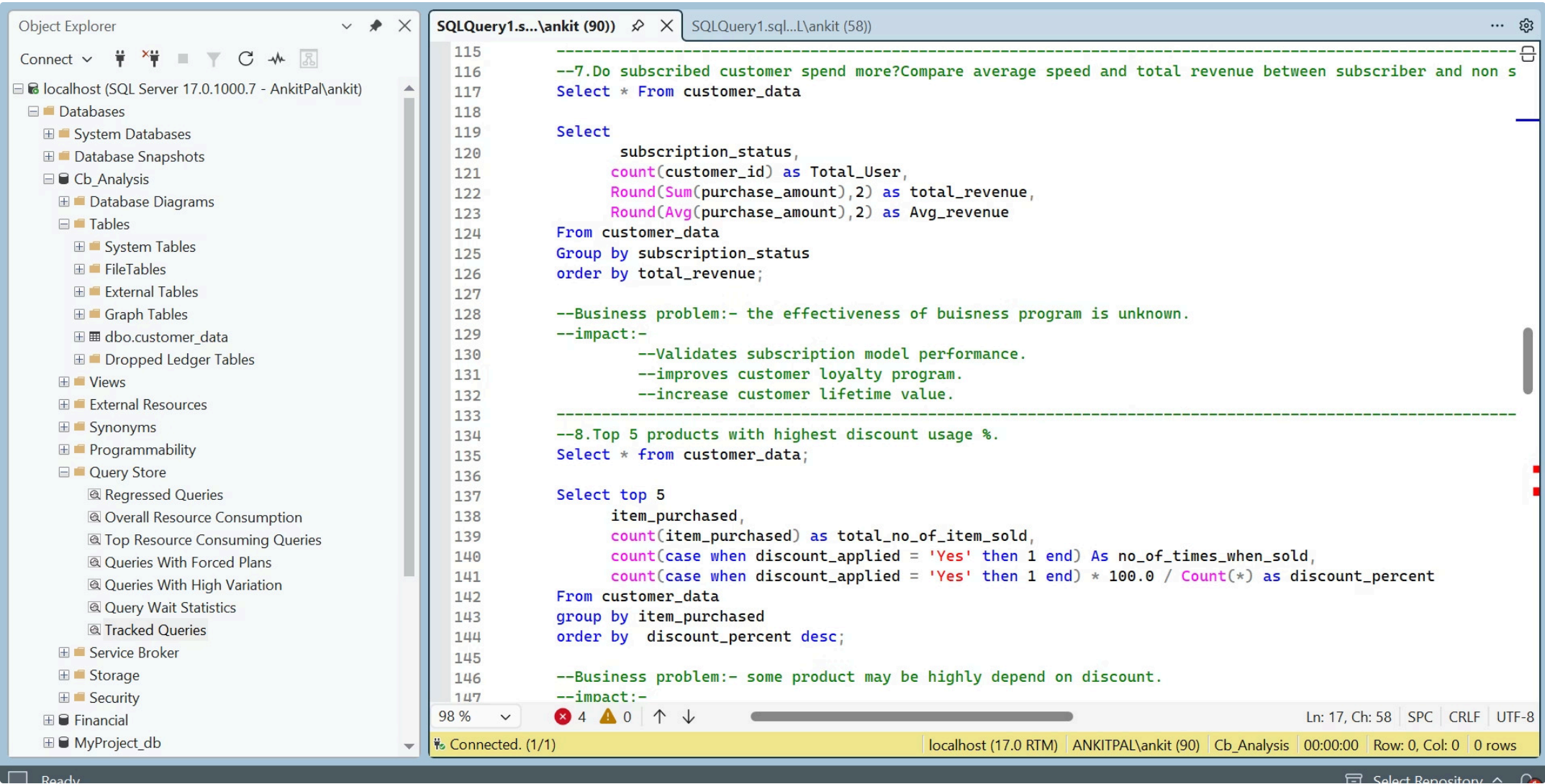
Discounts Work

Avg spend \$219 with discount vs. \$182 without

Male Leads

\$391K vs. Female \$352K vs. Other \$251K

SQL Analysis: Customer Segments & Subscriptions



Subscription Impact

Status	Customers	Avg Spend
Subscribed	1,579	\$270.58
Not Subscribed	3,421	\$165.72

✔ Subscribers spend **63% more** on average than non-subscribers.

51+ Age Group

Contributes **40.2%** of total revenue (\$399K) — the dominant segment.

SQL Analysis: Ratings & Top Products

```
--5.Which are top/bottom 5 products with the highest average review rating?

Select * From customer_data

Select Top 5
    item_purchased,
    Round(Avg(review_rating),2) As avg_rating
From customer_data
Group by item_purchased
Order by avg_rating Desc;

Select Top 5
    item_purchased,
    Round(Avg(review_rating),2) As avg_rating
From customer_data
Group by item_purchased
Order by avg_rating Asc;

--Business Problem:- No visibility into product performance based on customer satisfaction?
```

Top 5 by Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

Bottom 5 by Rating

Item	Avg Rating
Phone	2.94
Headphones	3.08
Watch	3.10
Bag	3.17
Laptop	3.19

Highest Discount Usage %

Item	Discount %
Laptop	51.7%
Phone	51.5%
Headphones	51.2%
Hat	50.0%
Watch	49.7%

⚠️ Electronics dominate discount usage — yet also have the **lowest ratings**.



Key Takeaways



Electronics Dominates

\$486K revenue — #1 category, but lowest customer ratings



Loyalty Pays Off

Subscribers spend 63% more; 3,085 loyal customers identified



Discounts Drive Value

Discounted purchases average \$219 vs. \$182 without



51+ Segment Rules

40.2% of total revenue — the highest-value age group